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ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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Mini Project Review-2 Presentation

AI-Powered Monitoring of Crop Health, Soil Condition, and Pest Risks Using Multispectral / Hyperspectral Imaging and Dataset-Based Analysis

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Abstract

CropSense AI is a dataset-based, AI-powered platform for monitoring crop health, vegetation stress, and pest risk without any physical sensors or IoT hardware. It combines three independently trained modules: a transfer-learning CNN (MobileNetV2) that classifies crop diseases across 14 species using the PlantVillage dataset, an NDVI-based vegetation stress classifier computed from multispectral Red and Near-Infrared imagery (Agriculture-Vision dataset), and an LSTM model that predicts pest-risk trends from historical weather and pest-survey data (ICAR-CRIDA, cotton).

As is expected for a dataset-based mini-project, no single real farm links all three signals together, so the three modules' outputs are combined at the output level into one weighted composite health score, delivered through a dashboard. All datasets and access methods have been feasibility-verified, and the disease-detection module has already been benchmarked on an independent real-world image set (PlantDoc), directly measuring the gap between lab-condition training and real-world performance.

Introduction

Agriculture is the backbone of India's economy, employing 46% of the workforce. Yet farmers face compounding challenges — unpredictable pest outbreaks, soil degradation, and climate variability — with minimal access to real-time, data-driven tools.

Precision agriculture addresses these challenges by using remote sensing and AI to enable early, accurate detection of crop stress. Multispectral imaging captures spectral bands beyond visible light, enabling vegetation indices like NDVI that directly correlate with plant health. Hyperspectral imaging extends this with hundreds of narrow bands for even finer analysis.

This project uses the Agriculture-Vision multispectral dataset for NDVI-based crop health, the PlantVillage and PlantDoc image datasets (54,000+ and 2,500+ images respectively) for disease detection, and pre-recorded pest-and-weather survey data from ICAR-CRIDA to build an end-to-end, dataset-only AI platform — no physical hardware required.

Motivation

Key Motivations for This Project:

1. Crop Loss: FAO reports 20–40% of global crop yield is lost annually to pests and diseases, costing the global economy over $220 billion.

2. Accessibility Gap: fewer than 20% of Indian farmers actively use digital agricultural technologies (Fortune India, 2025), and NITI Aayog's 2025 agri-tech roadmap identifies affordability and connectivity as the primary adoption barriers.

3. Late Detection: Traditional visual inspection detects crop stress only after visible symptoms — spectral imaging can detect stress earlier, before symptoms appear.

4. Market Opportunity: the global precision agriculture market is projected to grow from roughly $13 billion (2026) to over $22 billion within the next decade (Fortune Business Insights).

5. Software-First Approach: Using open-access datasets and free ML frameworks, this system eliminates expensive hardware requirements, making precision agriculture accessible without IoT investment.

Literature Survey

Base papers (one per module, IEEE Access / IEEE JSTARS, 2020–2026):

[1] Shrotriya et al., "An Approach Toward Classifying Plant-Leaf Diseases and Comparisons With the Conventional Classification," IEEE Access, 2024 — compares CNN-based approaches for leaf disease classification; base reference for the Module 1 CNN.

[2] Le, Liou & Pham, "Crop Response to Disease and Water Scarcity Quantified by Normalized Difference Latent Heat Index," IEEE Access, 2023 — quantifies crop stress via a normalized vegetation/heat index; base reference for the Module 2 NDVI scoring.

[3] Dong et al., "Automatic System for Crop Pest and Disease Dynamic Monitoring and Early Forecasting," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 13, pp. 4410–4418, 2020 — establishes a pest/disease early-forecasting framework; base reference for the Module 3 LSTM.

Supporting context: Kamilaris & Prenafeta-Boldú (2018) survey deep learning in agriculture broadly; Mohanty et al. (2016) establish CNN-based plant disease detection as a viable approach at scale.

Research Gaps

- No existing IEEE / Elsevier / Springer paper fuses disease, vegetation-index, and pest-risk signals into one system — the gap this project's fusion approach targets.

- Public datasets for each module come from different crops, farms, and countries — no dataset links all three signals to one real field. Addressed via output-level score fusion rather than raw-data fusion.

- Real-world generalization gap confirmed empirically: a lab-trained disease classifier dropped from 79.4% (lab images) to 18.1% (real-world PlantDoc images) in feasibility testing.

- Hyperspectral crop-disease datasets are scarce; the only feasible public option is wheat-specific, creating a crop mismatch with the multi-species disease module — being evaluated as a standalone demonstration.

- Pest-count field records are inherently sparse (real surveys are not weekly) — confirmed in the ICAR-CRIDA data (best case ~46% of weeks populated), requiring an explicit interpolation strategy.

- Whether hyperspectral imaging is a strict requirement of the problem statement, or whether the multispectral component alone would suffice, is still pending clarification with the project guide.

Problem Statement

Problem Statement :

Theme: Agriculture, FoodTech & Rural Development.

Problem: Farmers and agronomists lack real-time, precise tools for early detection of crop stress, soil degradation, and pest outbreaks. Traditional methods are reactive, labour-intensive, and inaccurate, leading to reduced yields and increased costs.

Proposed Solution: An AI-driven, dataset-based software platform that:

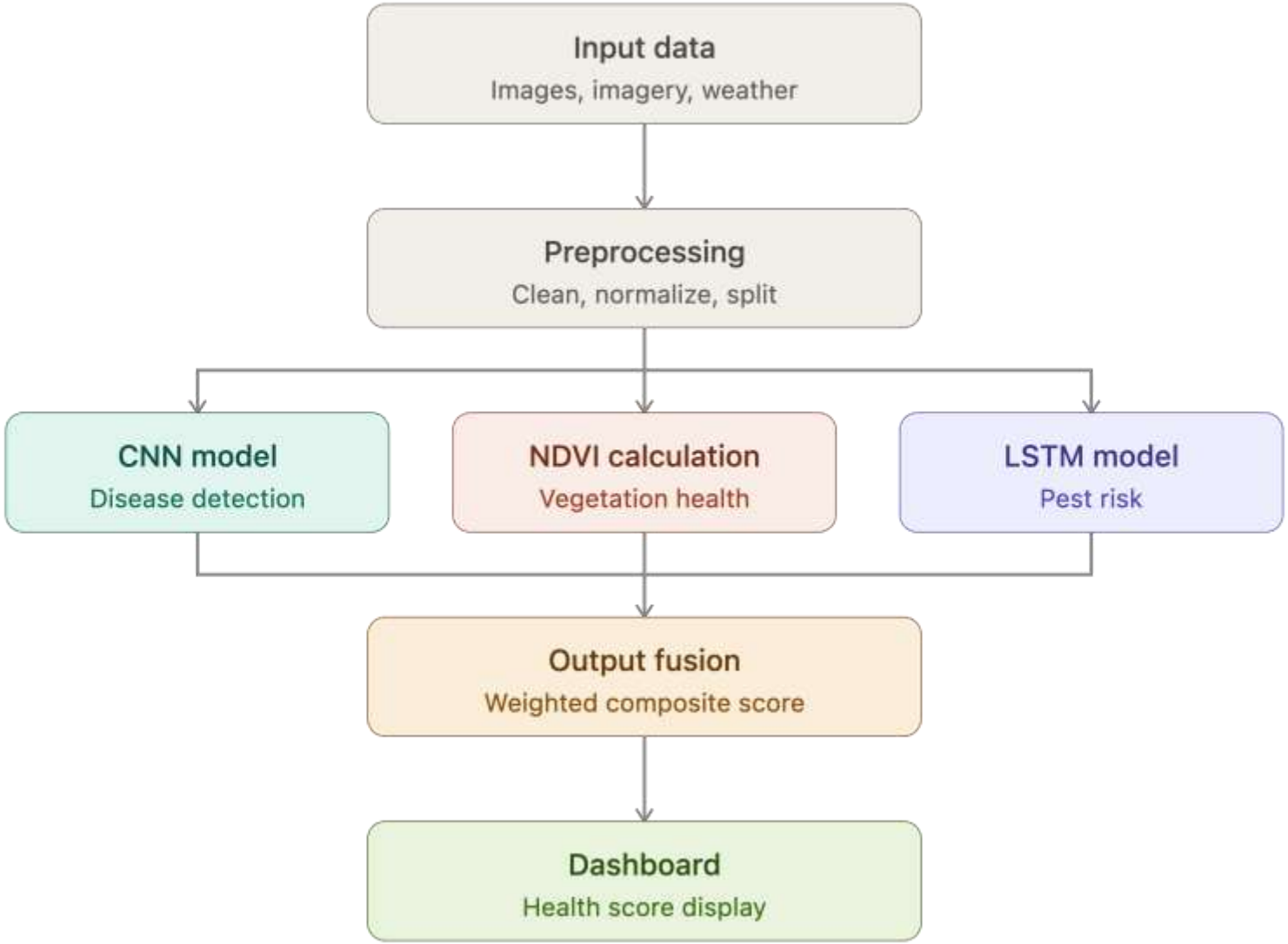
- Classifies crop diseases from images using a CNN (MobileNetV2 transfer learning) trained on the PlantVillage dataset across 14 crop species
- Computes NDVI vegetation stress categories from Agriculture-Vision multispectral (Red + NIR) imagery
- Predicts pest-risk trends using an LSTM model on historical pest-and-weather survey data (ICAR-CRIDA, cotton)
- Fuses all three module outputs into a single weighted composite health score
- Delivers results through a dashboard with health scores and recommendations

Scope (Dataset-Only): No physical sensors or IoT hardware. All data is sourced from public datasets and government agricultural research repositories.

Objectives

1. Build a disease-classification CNN generalizable across multiple crop species using transfer learning.
2. Compute NDVI-based vegetation stress categories from multispectral imagery.
3. Model pest-risk trends from historical weather and pest-survey time series using an LSTM.
4. Combine the three independent module outputs into one interpretable composite crop-health score.
5. Quantify real-world generalization by testing the trained disease model against an independent, non-lab image set.
6. Present results through an accessible, dataset-only software dashboard requiring no physical hardware.

Methodology — Architecture & Datasets



Methodology (contd.) — Implementation & Results

- Module 1: MobileNetV2 transfer learning, two-phase training (frozen → fine-tune), class weighting for a confirmed 36.2× class imbalance, augmentation (flip / rotate / zoom / contrast)

- Feasibility baseline (tomato subset): 79.4% accuracy on lab images, dropping to 18.1% on real-world PlantDoc images — confirms a genuine generalization gap; full 38-class model now in training

- Module 3: automated data-extraction pipeline built for ICAR-CRIDA (Python + requests + pandas), verified across 12 stations × 9 pests; Akola (Jassid, Aphid) selected as the primary training set (5 contiguous years, ~46% pest-record fill rate)

- Fusion formula (defined): composite_score = ⅓·disease_score + ⅓·NDVI_score + ⅓·(1 − pest_risk_score), equally weighted since no real dataset links all three signals to validate optimal weights

References (IEEE Format)

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[3] Y. Dong, F. Xu, L. Liu, X. Du, B. Ren, A. Guo, Y. Geng, C. Ruan, H. Ye, W. Huang, and Y. Zhu, "Automatic System for Crop Pest and Disease Dynamic Monitoring and Early Forecasting," IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, vol. 13, pp. 4410–4418, 2020.

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[5] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," Front. Plant Sci., vol. 7, p. 1419, Sep. 2016.

[6] T. Adão et al., "Hyperspectral imaging: A review on UAV-based sensors, data processing and applications for agriculture," Remote Sens., vol. 9, no. 11, p. 1110, Nov. 2017.

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Thank You

Queries and suggestions are welcome.